



Sri Lanka Institute of Information Technology

B. Sc. (Hons) in Information Technology

Specializing

Data Science

Final Examination

Year 3, Semester I (2018)

IT3031-Database Systems and Data-driven
Application

Duration: 2 Hours

June 2018

Instructions to Candidates:

- ◆ This paper is preceded by a 10-minute reading period. The supervisor will indicate when answering may commence.
- ◆ Write answers in the booklet given.
- ◆ The paper contains 4 questions. Answer All Questions.
- ◆ Total marks for the paper is 100.
- ◆ The paper contains 5 pages with the Cover Page.
- ◆ Electronic devices capable of storing and retrieving text, including calculators and mobile phones are not allowed.

Question 1

(25 marks)

Consider the following XML document that contains information on a collection of books.

```
'<?xml version="1.0" encoding="UTF-8"?>
<planes>
  <plane>
    <year> 1977 </year>
    <make> Cessna </make>
    <model> Skyhawk </model>
    <color> Light blue and white </color>
  </plane>
  <plane>
    <year> 1975 </year>
    <make> Piper </make>
    <model> Apache </model>
    <color> White </color>
  </plane>
  <plane>
    <year> 1960 </year>
    <make> Cessna </make>
    <model> Centurian </model>
    <color> Yellow and white </color>
  </plane>
  <plane>
    <year> 1956 </year>
    <make> Piper </make>
    <model> Tripacer </model>
    <color> Blue </color>
  </plane>
</planes>')
```

Note that the above XML document stored in *Planes(xText XML)* table created in MS SQL Server and it contains single record.

1. Compare XML data model verses the relational data model.
(4 marks)
2. Write an XQuery (FLWOR) expression to retrieve the models made before 1970.
Produce the following output.

```
<oldPlanes>
  <make> Cessna </make>
  <model> Centurian </model>
  <make> Piper </make>
  <model> Tripacer </model>
</oldPlanes>
```

(8 marks)

3. What would be the output of the following XQuery expression?

```
SELECT xText.query (' let $planes := /planes/plane
return <results>
{
  for $x in $planes
  where $x/year >= 1970
  order by ($x/year)[1]
```

```

        return ($x/make, $x/model,$x/year )
    }
</results>
')
FROM planes

```

(5 marks)

4. Write an XQuery expression to extract data from books XML document and create an HTML table containing the model of all the planes along with their color. Use the following HTML tags in your answer.

```

<table>
  <tr>
    <th>Model</th>
    <th>Color</th>
  </tr>
  ...
  <tr>
    <td> Tripacer </td>
    <td> Blue </td>
  </tr>
</table>

```

(8 marks)

Question 2

(25 marks)

1. If selections are performed before joins, the cost of executing the query is lower in most cases. Why?

(4 marks)

2. Consider the following relational schema:

```

Student(sno, sname, gpa)
Registration(sno, cno, marks)
Course(cno, cname, credits)

```

Assume that Student relation consists of 300 pages with 100 tuples per page, Registration relation contains 900 pages with 100 tuples per page, and Course relation contains 300 pages with 100 tuples per page. Assume equal-size fields in each relation and uniform distribution for Registration relation. Assume that 50 buffer pages are available.

Consider the following query:

```

Select      count(*)
From        Student s, Registration r
Where       s.sno = r.sno

```

- (a.) Describe the cost of performing the above query using the following join algorithms:

- (i.) Block-nested loop join

(5 marks)

- (ii.) Sort Merge Join

(5 marks)

(iii.) Index-nested loop join

Assume that Unclustered B+ tree index available on *Registration*<*sno*> with height 3. You may ignore the cost of storing *rid* in the leaf pages.

(5 marks)

(b.) Assume that the above join operation is very important and you are allowed to create any additional indexes. What indexes would you create? Explain your answer.

(6 marks)

Question 3

(25 marks)

1. Modern high performance database systems allow buffer frames to be stolen from uncommitted transactions. What problem does this create for the DBMS?
(4 marks)
2. In the context of a DBMS, explain what a database log is. In your answer, you should say what a log entry contains.
(5 marks)
3. What is the *Write-Ahead Logging* protocol?
(5 marks)
4. What is a *transaction table*, where is it stored, and what does it record?
(4 marks)
5. Briefly explain *UNDOing* and *REDOing* phases of crash recovery in DBMS.
(7 marks)

Question 4

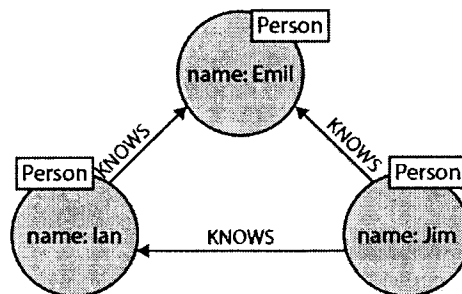
(25 marks)

1. (a.) Compare and Contrast RDBMS and NoSQL databases.
(2 marks)
(b.) Briefly explain the four types of data models in NoSQL databases.
(3 marks)
(c.) Explain the CAP theorem and BASE concept.
(2 marks)
2. (a.) What is a Collection in MongoDB?
(2 marks)
(b.) State how you can perform multiple updates to data in MongoDB.
(2 marks)

- (c.) Write the syntax to insert the following record. Assume that the collection name is movies.

```
{
  movie_id : "98020",
  title : "Ice Age Collision Course",
  director : "Mike Thurmeier",
  producer : "Lori Forte",
  duration : 94,
  released_year : 2016,
}
```

3. (a.) Explain three advantages of graph databases. (3 marks)
- (b.) Name the building blocks of a graph model. (2 marks)
- (c.) Name the building blocks of a graph model. (2 marks)
4. Following questions are related to Cypher Query Language (CQL).
- (a.) Explain MATCH and RETURN clauses in CQL. (2 marks)
- (b.) Name another three clauses that you can use with CQL queries. (2 marks)
- (c.) Write CQL queries to create the small social network represented below. (2 marks)



(3 marks)

- End of the Examination Paper -